

Task 3 - Wireshark Packet Analysis Report

Network Interfaces Monitored: vboxnet0 , wlan1

Protocols Analyzed: HTTP, DNS, TCP

Tools Used: Wireshark for packet analysis & Hping3 to generate network anomalies

1. Objective

The objective of this analysis was to capture and examine network traffic to understand HTTP, DNS, and TCP behaviors. Key tasks included:

- Capturing HTTP and DNS requests/responses
- Observing TCP handshake sequences
- Identifying any unusual or malformed packets

2. Methodology

Network traffic was captured using Wireshark on the interfaces vboxnet0 and wlan1 . Filters were applied to isolate relevant traffic:

- HTTP: http
- DNS: dns
- TCP port 80: tcp.port==80

Each packet was analyzed for:

- Source and destination IPs
- Protocol-specific headers and flags
- Sequence and acknowledgment numbers
- Anomalies such as repeated resets or malformed packets

3. Observations

3.1 HTTP Traffic

- **Captured Traffic:** GET and HEAD requests from 192.168.56.1 to 192.168.56.101 .
- **Details:**
 - HEAD / HTTP/1.1 request using curl/8.19.0 as the User-Agent.
 - HTTP response returned 200 OK , indicating successful communication.
- **Conclusion:** Normal HTTP communication. No irregular headers or unexpected behavior observed.

Wireshark HTTP traffic result:

Capturing from vboxnet0

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h

http

No.

Time

Source

Destination

Protocol

Length Info

7

0.010250983

192.168.56.1

192.168.56.101

HTTP

145 HEAD / HTTP/1.1

9

0.049694212

192.168.56.101

192.168.56.1

HTTP

221 HTTP/1.1 200 OK

▼ Frame 7: Packet, 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits) on interface vboxnet0, id 0

Section number: 1

Interface id: 0 (vboxnet0)

Encapsulation type: Ethernet (1)

Arrival Time: Apr 25, 2026 12:07:51.585705015 +02

UTC Arrival Time: Apr 25, 2026 10:07:51.585705015 UTC

Epoch Arrival Time: 177711071.585705015

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 147.032 microseconds]

[Time since reference or first frame: 10.250983 milliseconds]

Frame Number: 7

Frame Length: 145 bytes (1160 bits)

Capture Length: 145 bytes (1160 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:tcp:http]

Character encoding: ASCII (0)

[Coloring Rule Name: HTTP]

[Coloring Rule String: http || tcp.port == 80 || http2]

Ethernet II, Src: 0a:00:27:00:00:00 (0a:00:27:00:00:00), Dst: PCSSystemtec_ba:8d:b6 (08:00:27:ba:8d:b6)

Destination: PCSSystemtec_ba:8d:b6 (08:00:27:ba:8d:b6)

Source: 0a:00:27:00:00:00 (0a:00:27:00:00:00)

Type: IPv4 (0x0008)

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.101

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 131

Identification: 0xb3fc (46076)

010 = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to live: 64

Protocol: TCP (6)

Header Checksum: 0x94c1 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.56.1

Destination Address: 192.168.56.101

[Stream index: 0]

Transmission Control Protocol, Src Port: 52558, Dst Port: 80, Seq: 1, Ack: 1, Len: 79

Hypertext Transfer Protocol

HEAD / HTTP/1.1\r\n

Host: 192.168.56.101\r\n

User-Agent: curl/8.19.0\r\n

Accept: */*\r\n

\r\n

[Response in frame: 9]

[Full request URI: http://192.168.56.101/]

0000 08 00 27 ba 8d b6 0a 00 27 00 00 00 00 00 45 00E..

0010 00 83 b3 fc 40 00 40 06 94 c1 c0 a8 38 01 c0 a88...

0020 38 65 cd 4e 00 50 d1 53 4b af f4 c1 e7 68 80 18 8e N P S K ...h...

0030 00 3f f2 2c 00 00 01 01 08 0a 7c 5a 46 9d 00 01 ?[ZF...

0040 ce b9 48 45 41 44 20 2f 20 48 54 54 50 2f 31 2e ..HEAD / HTTP/1.

0050 31 0d 0a 48 6f 73 74 3a 20 31 39 32 2e 31 36 38 1 Host: 192.168

0060 2e 35 36 2e 31 30 31 0d 0a 55 73 05 72 2d 41 67 .56.101 User-Ag

0070 65 6e 74 3a 20 63 75 72 6c 2f 38 2e 31 39 2e 30 ent: cur l/8.19.0

0080 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0d -Accept: */*...

0090 0a

● Hypertext Transfer Protocol: Protocol

Packets: 15 • Displayed: 2 (13.3%)

Profile: Default

3.2 DNS Traffic

- **Captured Traffic:** DNS queries and responses with the server 8.8.8.8 .
- **Query Example:** netflix.com
- **Response:** Multiple IPv4 addresses returned:
 - 54.73.148.110
 - 54.155.246.232
 - 18.200.8.190
- **Observation:** Standard DNS resolution. No evidence of spoofed or manipulated responses.

Wireshark DNS traffic result:

Capturing from wlan1

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d

dns

No.

Time

Source

Destination

Protocol

Length Info

59

21.641876565

192.168.29.32

8.8.8.8

DNS

71 Standard query 0xcd63 A netflix.com

60

21.691094500

8.8.8.8

192.168.29.32

DNS

119 Standard query response 0xcd63 A netflix.com A 54.73.148.110 A 54.155.246.232 A 18.200.8.190

61

21.711386048

192.168.29.32

8.8.8.8

DNS

71 Standard query 0x7624 AAAA netflix.com

62

21.914041610

8.8.8.8

192.168.29.32

DNS

155 Standard query response 0x7624 AAAA netflix.com AAAA 2a05:d018:76c:b684:b233:ac1f:be1f:7 AAAA 2a05:d018:76c:b685:c898:aa3a:42c7:9d2...

▼ Frame 60: Packet, 119 bytes on wire (952 bits), 119 bytes captured (952 bits) on interface wlan1, id 0

Ethernet II, Src: Servercomprij_ed:3f:8f (a8:da:0c:ed:3f:8f), Dst: a2:ee:28:b1:bd:48 (a2:ee:28:b1:bd:48)

Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.29.32

User Datagram Protocol, Src Port: 53, Dst Port: 44718

Domain Name System (response)

Transaction ID: 0xcd63

Flags: 0x0180 Standard query response, No error

Questions: 1

Answer RRs: 3

Authority RRs: 0

Additional RRs: 0

Queries

netflix.com: type A, class IN

Name: netflix.com

[Name length: 11]

[Label Count: 2]

Type: A (1) (Host Address)

class: IN (0x0001)

Answers

netflix.com: type A, class IN, addr 54.73.148.110

netflix.com: type A, class IN, addr 54.155.246.232

netflix.com: type A, class IN, addr 18.200.8.190

[Request in: 59]

[Time: 67.588915 milliseconds]

0000 a2 ee 28 b1 bd 48 a8 da 0c ed 3f 8f 08 00 45 00H...?..E..

0010 00 69 c6 f7 00 00 0b 11 9a b4 08 08 08 00 c0 a8k.....

0020 1d 20 00 35 ae ae 00 55 b3 b3 cd 63 81 80 00 015..U.....

0030 00 03 00 00 00 00 07 6e 65 74 66 6c 69 78 03 63etflix.c...

0040 6f dd 00 00 01 00 01 c0 0c 00 01 00 01 00 00 00 om.....

0050 3c 00 04 36 49 94 6e c0 0c 00 01 00 01 00 00 00 < 6I.n.....

0060 3c 00 04 36 9b f6 e8 c0 0c 00 01 00 01 00 00 00 < 6.....

0070 3c 00 04 12 c8 08 be 0c 00 01 00 01 00 00 00 <

● Domain Name System: Protocol

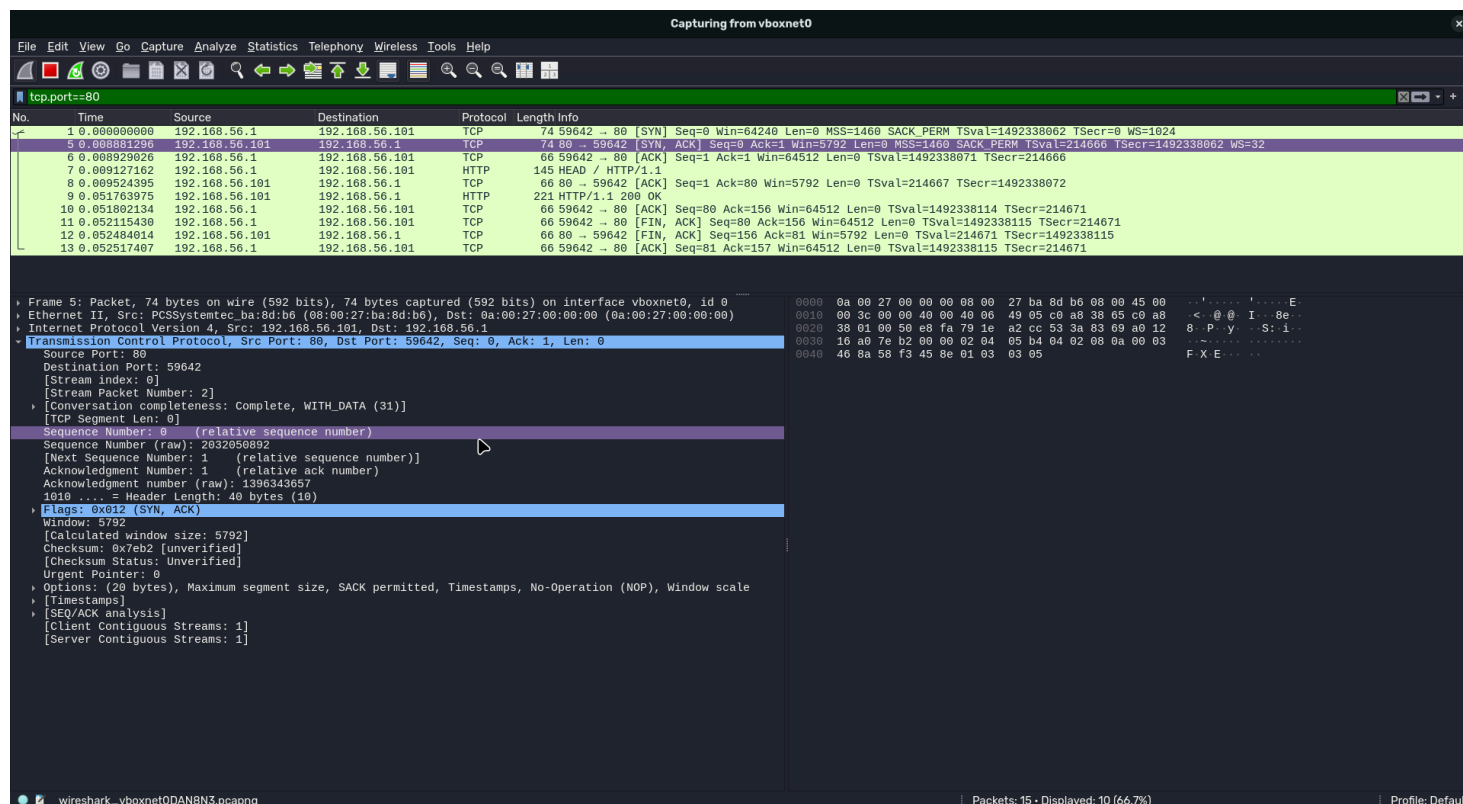
Packets: 265 • Displayed: 4 (1.5%)

Profile: Default

3.3 TCP Handshakes

- **Procedure:** Monitored TCP three-way handshake for port 80 connections.
- **Sequence:** SYN → SYN-ACK → ACK completed successfully.
- **Flags Observed:**
 - SYN packets: 0x002
 - SYN-ACK packets: 0x012
- **Observation:** TCP handshake and connection setup occurred as expected, confirming normal session initiation.

Wireshark TCP traffic result:



3.4 Unusual Packets – SYN Flood

- **Observation:** Wireshark captured multiple TCP Port numbers reused and RST packets on port 80.
- **Cause:** These packets were generated by the command: `hping3 -S --flood -V -p 80 192.168.56.101`
- `-S` sets the SYN flag.
- `--flood` sends packets as fast as possible.
- `-V` enables verbose output.
- `-p 80` targets TCP port 80.
- Destination IP: 192.168.56.101 (Metasploitable2 VM).
- **Analysis:** The SYN packets flood the target machine, producing rapid SYN-ACK responses, followed by resets (RST) as the ports are reused. This simulates a SYN flood attack in a controlled lab environment.
- **Conclusion:** The unusual packets are expected and intentional. They do not indicate an actual network compromise, but rather serve as part of testing and experimentation.

Conclusion

The network traffic captured and analyzed demonstrates normal operations for HTTP, DNS, and TCP protocols under typical conditions. HTTP requests and responses were complete and correctly formatted, DNS queries resolved accurately to multiple IP addresses, and TCP handshakes were properly executed with expected sequence and acknowledgment numbers.

The observed anomalies—specifically the rapid sequence of SYN packets and associated RST responses on port 80—were the result of an intentional SYN flood generated using `hping3` against the Metasploitable2 VM. These packets were part of a controlled experiment to simulate high-volume traffic and did not represent a security threat or network malfunction.

Overall, the traffic patterns align with expected behavior in a laboratory environment, and all unusual packets were accounted for and intentionally induced. The analysis confirms both the proper functioning of standard network protocols and the effectiveness of packet capture and inspection using Wireshark.